LAB 7- From anomaly detection to incidents

Lab Task 1 – Generalize your Lab 6 monitor to track any service by K8S deployment. configuration using the same docker image.

I provisioned my cluster using the provided provisioned script with the name as shown.

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ ./provision2.sh onlineboutique aiops4 europe-west4

No current clusters found, continuing to deploy one
Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. S
bernetes-engine/docs/how-to/disable-kubelet-readonly-port for ways to check usage and for migration instructions.
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster onlineboutique in europe-west4-a... Cluster is being health-checked (Kubernetes Control Plane is healthy)...d
Created [https://container.googleapis.com/v1/projects/aiops4/zones/europe-west4-a/clusters/onlineboutique].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload_/gcloud/europe-west4-a/on
kubeconfig entry generated for onlineboutique.
NAME            LOCATION        MASTER_VERSION       MASTER_IP     MACHINE_TYPE   NODE_VERSION         NUM_NODES  STATUS
onlineboutique  europe-west4-a  1.30.5-gke.1699000   34.90.116.92  e2-standard-4  1.30.5-gke.1699000   1          RUNNING
```

Dockerfile for the anomaly-moniotor.py

```
Dockerfile
monitorapp > Dockerfile > ...
FROM python:3.9-slim
RUN apt-get update && apt-get install -y \
    build-essential \
    python3-dev \
    gcc \
    && rm -rf /var/lib/apt/lists/*
WORKDIR /app
RUN mkdir -p /app/training
COPY boutique_train.json /app/training/frontend_shippingservice.json
COPY boutique_train.json /app/training/frontend_productcatalog.json
COPY anomaly-monitor.py /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
EXPOSE  8001 8002
ENV PYTHONUNBUFFERED=1
CMD ["python", "anomaly-monitor.py"]
```

Building the docker image

```
oadewusi@oluSec:~/aiops/Assignments/LAB7/monitorapp$ docker build -t oadewusi/anomaly-monitor:v4 .
[+] Building 1.8s (14/14) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                            0.0s
 => => transferring dockerfile: 546B                                                                            0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                              1.7s
 => [internal] load .dockerignore                                                                               0.0s
 => => transferring context: 2B                                                                                 0.0s
 => [1/9] FROM docker.io/library/python:3.9-slim@sha256:4ee0613170ac55ebc693a03b6655a5c6f387126f6bc3390e739c2e6c337880ef  0.0s
 => [internal] load build context                                                                               0.0s
 => => transferring context: 114B                                                                               0.0s
 => CACHED [2/9] RUN apt-get update && apt-get install -y     build-essential     python3-dev     gcc     && rm -rf /var/lib/a  0.0s
 => CACHED [3/9] WORKDIR /app                                                                                   0.0s
 => CACHED [4/9] RUN mkdir -p /app/training                                                                     0.0s
 => CACHED [5/9] COPY boutique_train.json /app/training/frontend_shippingservice.json                           0.0s
 => CACHED [6/9] COPY boutique_train.json /app/training/frontend_productcatalog.json                            0.0s
 => CACHED [7/9] COPY anomaly-monitor.py /app                                                                   0.0s
```

Pushing the image to Docker hub

```
oadewusi@oluSec:~/aiops/Assignments/LAB7/monitorapp$ docker push oadewusi/anomaly-monitor:v4
The push refers to repository [docker.io/oadewusi/anomaly-monitor]
f88460320de6: Layer already exists
36aa4c9d8861: Layer already exists
bc103d96e777: Layer already exists
005c2809b286: Layer already exists
45a3698a16a2: Layer already exists
b60405d2bcb5: Layer already exists
8db6237317d0: Layer already exists
07cc7ecf9d04: Layer already exists
639cbe945cf9: Layer already exists
13f3346e3d1d: Layer already exists
0fd5a9e94c37: Layer already exists
```

Deployment yaml file for frontend to shipping.

frontend_shipping.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: anomaly-monitor-frontend-shipping
spec:
  replicas: 1
  selector:
    matchLabels:
      app: anomaly-monitor-frontend-shipping
  template:
    metadata:
      labels:
        app: anomaly-monitor-frontend-shipping
      annotations:
        prometheus.io/scrape: "true"
        prometheus.io/port: "8001"
        prometheus.io/path: "/metrics"
    spec:
      containers:
      - name: anomaly-monitor
        image: index.docker.io/oadewusi/anomaly-monitor:v4
        imagePullPolicy: Always
        command: ["python"]
        args:
          [
            "anomaly-monitor.py",
            "-f", "frontend",
            "-t", "shippingservice",
            "-d", "/app/training/frontend_shippingservice.json",
```

Deployment yaml file for frontend to productcatalogservice.

frontend_productcatalog.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: anomaly-monitor-frontend-productcatalog
spec:
  replicas: 1
  selector:
    matchLabels:
      app: anomaly-monitor-frontend-productcatalog
  template:
    metadata:
      labels:
        app: anomaly-monitor-frontend-productcatalog
      annotations:
        prometheus.io/scrape: "true"
        prometheus.io/port: "8002"
        prometheus.io/path: "/metrics"
    spec:
      containers:
      - name: anomaly-monitor
        image: index.docker.io/oadewusi/anomaly-monitor:v4
        imagePullPolicy: Always
        command: ["python"]
        args:
          [
            "anomaly-monitor.py",
            "-f", "frontend",
            "-t", "productcatalogservice",
            "-d", "/app/training/frontend_productcatalog.json",
```

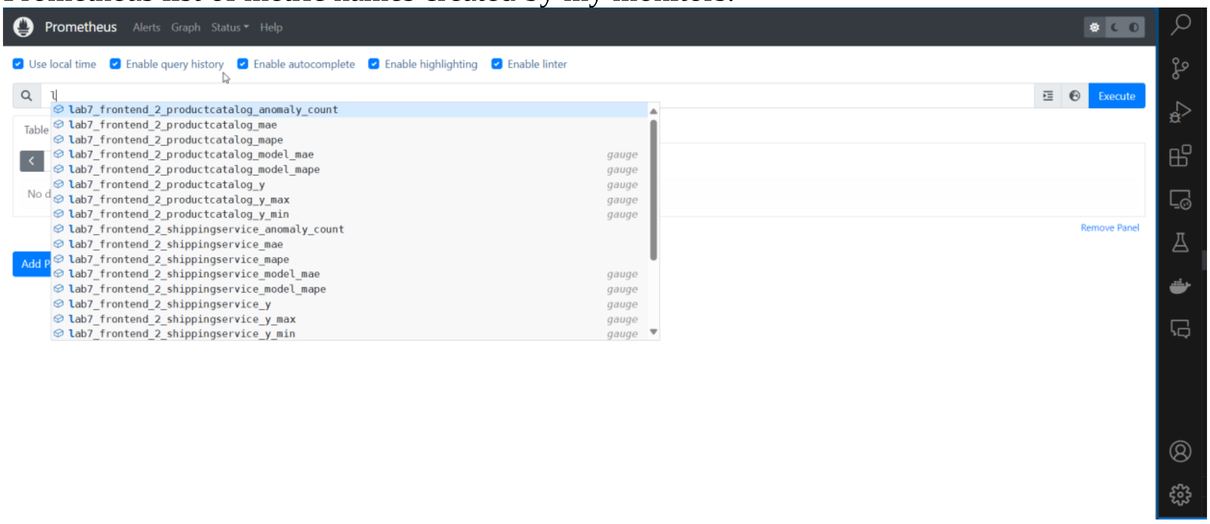
Screenshots of my anomaly-monitor.py

```python
import argparse
import time
import requests
import pandas as pd
import numpy as np
from prophet import Prophet
from datetime import datetime
import json
from prometheus_client import start_http_server, Gauge

parser = argparse.ArgumentParser()
parser.add_argument("-f", "--from_service", required=True, help="Source service name")
parser.add_argument("-t", "--to_service", required=True, help="Destination service name")
parser.add_argument("-d", "--training_data", required=True, help="Path to training data JSON file")
parser.add_argument("-p", "--port", required=True, type=int, help="Prometheus scrape port")
parser.add_argument("--prefix", required=True, help="Prefix for Prometheus metrics")
args = parser.parse_args()

# Prometheus metrics
prefix = args.prefix
anomaly_count_gauge = Gauge(f"{prefix}_anomaly_count", "Number of detected anomalies")
mae_gauge = Gauge(f"{prefix}_model_mae", "Mean Absolute Error of the model")
mape_gauge = Gauge(f"{prefix}_model_mape", "Mean Absolute Percentage Error of the model")
y_gauge = Gauge(f"{prefix}_y", "Observed value of y")
y_min_gauge = Gauge(f"{prefix}_y_min", "Predicted minimum value for y")
y_max_gauge = Gauge(f"{prefix}_y_max", "Predicted maximum value for y")

# Prometheus query
url = 'http://prometheus.istio-system:9090/api/v1/query'
query = f'histogram_quantile(0.5, rate(istio_request_duration_milliseconds_bucket{{source_app="{args.from_service}", destination_app="{args.to_service}", reporter="source"}}[1m]))'

def load_training_data():
    with open(args.training_data, 'r') as file:
        data = json.load(file)

```

Applying both deployment to the cluster

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl apply -f frontend_shipping.yaml
deployment.apps/anomaly-monitor-frontend-shipping created
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl apply -f frontend_productcatalog.yaml
deployment.apps/anomaly-monitor-frontend-productcatalog created
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl get pods
NAME                                                      READY   STATUS    RESTARTS   AGE
adservice-64586ccfb9-xlrpz                                2/2     Running   0          5m16s
anomaly-monitor-frontend-productcatalog-677b894f5f-wxs2w  2/2     Running   0          40s
anomaly-monitor-frontend-shipping-988b499b4-q6jj7         2/2     Running   0          51s
cartservice-6bd5c944b4-4tmq4                              2/2     Running   0          5m15s
checkoutservice-6d4bcd6f95-wn8tw                          2/2     Running   0          5m15s
currencyservice-f987888c-6xs7c                            2/2     Running   0          5m14s
emailservice-56fbcfbbf7-mjs8s                             2/2     Running   0          5m14s
frontend-cb9967686-m7plr                                  2/2     Running   0          5m13s
istio-gateway-istio-5b8576f55b-hmhgz                      1/1     Running   0          5m9s
loadgenerator-7446879c5-zgs4p                             2/2     Running   0          5m6s
paymentservice-794b5dfdf7-7smgg                           2/2     Running   0          5m12s
productcatalogservice-74b4c878d5-xjrqm                    2/2     Running   0          5m12s
recommendationservice-65dd5bd87c-2r8ls                    2/2     Running   0          5m11s
redis-cart-7ff8f4d6ff-cdslc                               2/2     Running   0          5m11s
```

Prometheus list of metric names created by my monitors.



Console log of running the monitor 1

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl logs -f anomaly-monitor-frontend-shipping-988b499b4-q6jj7
Importing plotly failed. Interactive plots will not work.
14:41:16 - cmdstanpy - INFO - Chain [1] start processing
14:41:16 - cmdstanpy - INFO - Chain [1] done processing

 ID        Timestamp           Anomalies    Observed     Predicted      MAE        MAPE
-------------------------------------------------------------------------------------------
 0     2024-12-10 14:41:16        1         3.000000     3.853515     0.853515    0.284505
 1     2024-12-10 14:42:17        0         3.000000     3.750629     0.750629    0.250210
 2     2024-12-10 14:43:17        0         3.000000     3.659448     0.659448    0.219816
 3     2024-12-10 14:44:17        0         3.000000     3.579353     0.579353    0.193118
 4     2024-12-10 14:45:17        0         3.000000     3.509722     0.509722    0.169907
 5     2024-12-10 14:46:17        0         3.000000     3.449925     0.449925    0.149975
 6     2024-12-10 14:47:17        0         3.000000     3.399326     0.399326    0.133109
 7     2024-12-10 14:48:17        0         3.000000     3.357286     0.357286    0.119095
 8     2024-12-10 14:49:17        0         3.000000     3.322653     0.322653    0.107551
 9     2024-12-10 14:50:18        0         3.076923     3.295904     0.218981    0.071169
 10    2024-12-10 14:51:18        0         3.000000     3.275753     0.275753    0.091918
 11    2024-12-10 14:52:18        0         3.086957     3.261543     0.174587    0.056556
```

Console log of running the monitor 2

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl logs -f anomaly-monitor-frontend-productcatalog-677b894f5f-wxs2w
Importing plotly failed. Interactive plots will not work.
14:41:16 - cmdstanpy - INFO - Chain [1] start processing
14:41:16 - cmdstanpy - INFO - Chain [1] done processing

 ID        Timestamp           Anomalies    Observed     Predicted      MAE        MAPE
-------------------------------------------------------------------------------------------
 0     2024-12-10 14:41:16        1         3.006873     3.853515     0.846642    0.281569
 1     2024-12-10 14:42:16        0         3.024779     3.750629     0.725850    0.239968
 2     2024-12-10 14:43:17        0         3.000000     3.659448     0.659448    0.219816
 3     2024-12-10 14:44:17        0         3.000000     3.579353     0.579353    0.193118
 4     2024-12-10 14:45:17        0         3.003781     3.509722     0.505942    0.168435
 5     2024-12-10 14:46:17        0         3.015414     3.449925     0.434511    0.144097
 6     2024-12-10 14:47:17        0         3.000000     3.399326     0.399326    0.133109
 7     2024-12-10 14:48:17        0         3.003165     3.357286     0.354121    0.117916
 8     2024-12-10 14:49:17        0         3.000000     3.322653     0.322653    0.107551
 9     2024-12-10 14:50:17        0         3.010152     3.295904     0.285752    0.094929
 10    2024-12-10 14:51:18        0         3.007435     3.275753     0.268318    0.089218
 11    2024-12-10 14:52:18        0         3.007663     3.261543     0.253880    0.084411
```

Lab Task 2 – Train and deploy anomaly monitors for 2 selected services.

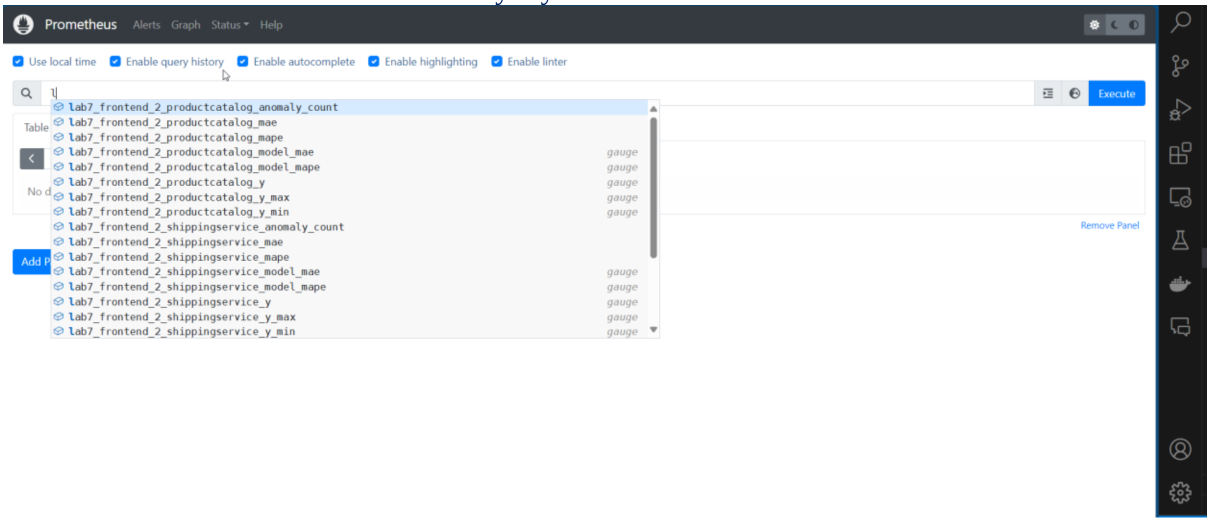
Console log of running the monitor 1


```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl logs -f anomaly-monitor-frontend-shipping-988b499b4-q6jj7
Importing plotly failed. Interactive plots will not work.
14:41:16 - cmdstanpy - INFO - Chain [1] start processing
14:41:16 - cmdstanpy - INFO - Chain [1] done processing

 ID        Timestamp           Anomalies    Observed     Predicted      MAE        MAPE
-------------------------------------------------------------------------------------------
 0     2024-12-10 14:41:16        1         3.000000     3.853515     0.853515    0.284505
 1     2024-12-10 14:42:17        0         3.000000     3.750629     0.750629    0.250210
 2     2024-12-10 14:43:17        0         3.000000     3.659448     0.659448    0.219816
 3     2024-12-10 14:44:17        0         3.000000     3.579353     0.579353    0.193118
 4     2024-12-10 14:45:17        0         3.000000     3.509722     0.509722    0.169907
 5     2024-12-10 14:46:17        0         3.000000     3.449925     0.449925    0.149975
 6     2024-12-10 14:47:17        0         3.000000     3.399326     0.399326    0.133109
 7     2024-12-10 14:48:17        0         3.000000     3.357286     0.357286    0.119095
 8     2024-12-10 14:49:17        0         3.000000     3.322653     0.322653    0.107551
 9     2024-12-10 14:50:18        0         3.076923     3.295904     0.218981    0.071169
 10    2024-12-10 14:51:18        0         3.000000     3.275753     0.275753    0.091918
 11    2024-12-10 14:52:18        0         3.086957     3.261543     0.174587    0.056556
```


Console log of running the monitor 2

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl logs -f anomaly-monitor-frontend-productcatalog-677b894f5f-wxs2w
Importing plotly failed. Interactive plots will not work.
14:41:16 - cmdstanpy - INFO - Chain [1] start processing
14:41:16 - cmdstanpy - INFO - Chain [1] done processing

  ID         Timestamp          Anomalies     Observed      Predicted        MAE          MAPE
-----------------------------------------------------------------------------------------------
  0      2024-12-10 14:41:16        1         3.006873      3.853515      0.846642      0.281569
  1      2024-12-10 14:42:16        0         3.024779      3.750629      0.725850      0.239968
  2      2024-12-10 14:43:17        0         3.000000      3.659448      0.659448      0.219816
  3      2024-12-10 14:44:17        0         3.000000      3.579353      0.579353      0.193118
  4      2024-12-10 14:45:17        0         3.003781      3.509722      0.505942      0.168435
  5      2024-12-10 14:46:17        0         3.015414      3.449925      0.434511      0.144097
  6      2024-12-10 14:47:17        0         3.000000      3.399326      0.399326      0.133109
  7      2024-12-10 14:48:17        0         3.003165      3.357286      0.354121      0.117916
  8      2024-12-10 14:49:17        0         3.000000      3.322653      0.322653      0.107551
  9      2024-12-10 14:50:17        0         3.010152      3.295904      0.285752      0.094929
  10     2024-12-10 14:51:18        0         3.007435      3.275753      0.268318      0.089218
  11     2024-12-10 14:52:18        0         3.007663      3.261543      0.253880      0.084411
```

Prometheus list of metric names created by my monitors.

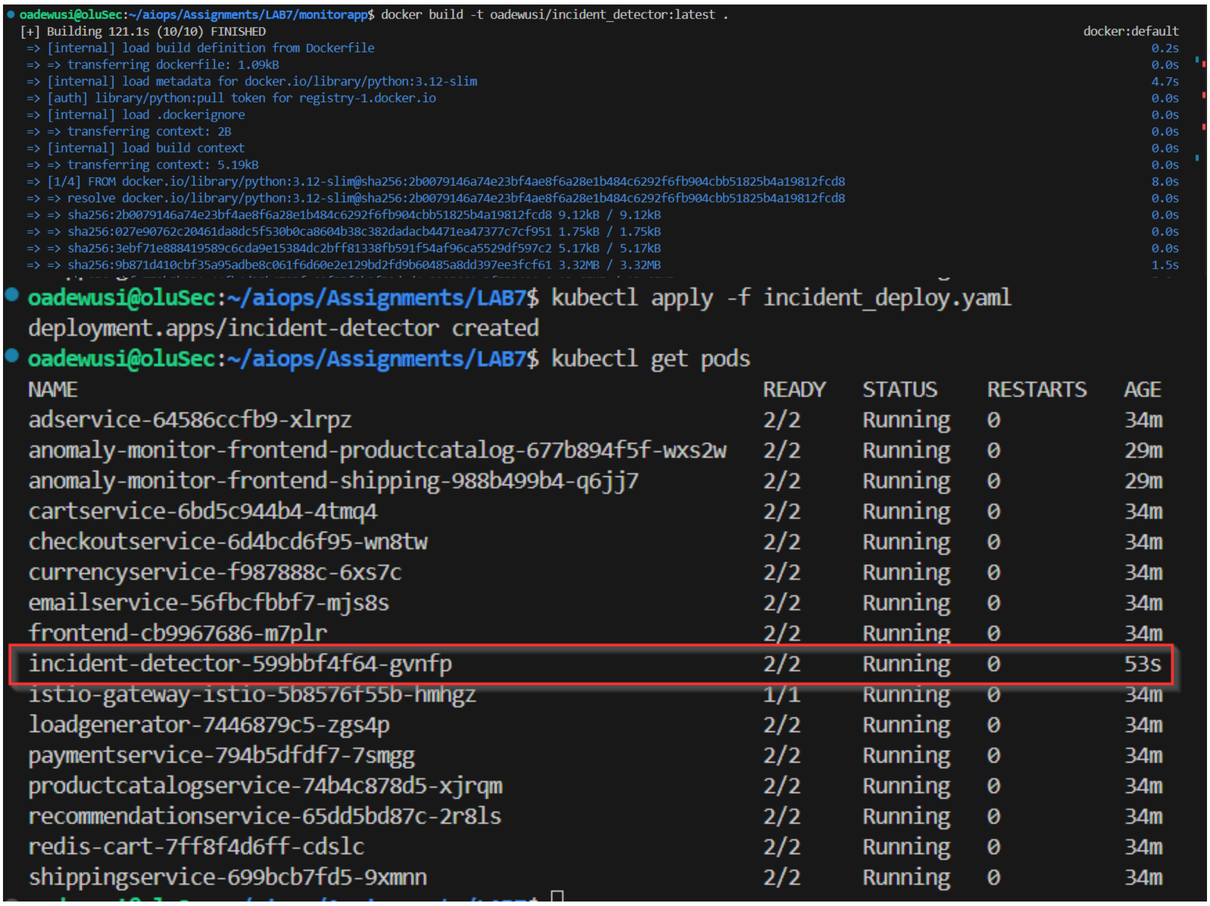


Lab Task 3 – Implement an Incident detection monitor consuming 2 anomaly metrics.

Code for the incident_detector.py

```
incident_detector.py
incidentapp > incident_detector.py
import argparse
import random

# Argument parsing
parser = argparse.ArgumentParser()
parser.add_argument("--threshold", type=int, default=4, help="Incident threshold")
parser.add_argument("--services", nargs=2, required=True, help="Two service prefixes (SVC1, SVC2)")
parser.add_argument("--port", type=int, required=True, help="Prometheus scrape port")
args = parser.parse_args()

# Prometheus metrics
temperature_gauge = Gauge("incident_temperature", "Current sum of accumulators")
sev1_gauge = Gauge("incident_sev1", "Sev 1 incident flag (0 or 1)")
sev2_gauge = Gauge("incident_sev2", "Sev 2 incident flag (0 or 1)")

# Initialize accumulators for the services
accumulators = {service: 0 for service in args.services}

# Prometheus query configuration
PROMETHEUS_URL = "http://prometheus.istio-system:9090/api/v1/query"
ANOMALY_QUERY_TEMPLATE = "sum({prefix}_anomaly_count) by (job)"


def fetch_anomaly_count(service):
    """
    Fetches the anomaly count for a given service using Prometheus queries.
    Implements retries with exponential backoff in case of failures.
    """
    query = ANOMALY_QUERY_TEMPLATE.format(prefix=service)
    retries = 3
    for attempt in range(retries):
        try:
            if attempt > 0:
                print(f"Retrying query for {service}: Attempt {attempt + 1}")
            else:
                print(f"Querying Prometheus for {service}: {query}")
```

Building docker image for the incident_detector.py



Explanation of how I refined the flowchart logic and why.

1. Initialization of Accumulators

- **Flowchart Step**: The accumulators for both services are initialized to zero at the beginning.
- **My Code Enhancements:** Dynamically initializes accumulators based on the --services argument:

```
accumulators = {service: 0 for service in args.services}
```

Reason: This makes the code generic and scalable, allowing it to monitor any two services without requiring code changes.

2. Handling Anomalies

- **Flowchart Step**: Checks if a service has an anomaly and updates the accumulator accordingly.
- **My Code Enhancements:** Implements anomaly handling with logic to increment or decrement accumulators:

```
if anomaly_count > 0: accumulators[service] += 1 else: accumulators[service] = max(0, accumulators[service] - 2)
```

Reason:

- Incrementing the accumulator (+1) ensures anomalies are registered.
- Decrementing the accumulator (-2) avoids false positives caused by transient issues, ensuring only persistent anomalies are considered for incidents.

3. Computing Total Temperature

- **Flowchart Step**: Calculates the total temperature (sum of accumulators) and compares it to a threshold.
- **My Code Enhancements:** Calculates the total temperature:

```
total_temperature = sum(accumulators.values())
temperature_gauge.set(total_temperature)
```

Reason: Total temperature serves as a key indicator for detecting incidents. Publishing it as a Prometheus gauge allows visualization in real-time on Grafana.

4. Determining Incident Severity

- **Flowchart Step:**
    - Declares a Sev 1 incident if both services have non-zero accumulators and the total temperature exceeds the threshold.
    - Declares a Sev 2 incident if the total temperature exceeds the threshold but not all services have non-zero accumulators.
- **My Code Enhancements:** Implements the severity logic:

```
sev1_flag = 1 if total_temperature > args.threshold and all(acc > 0 for acc in accumulators.values()) else 0
sev2_flag = 1 if total_temperature > args.threshold and not sev1_flag else 0
```

Reason:

- This logic ensures that Sev 1 incidents require anomalies in both services, indicating a higher severity problem.
- Sev 2 incidents are triggered for less severe cases, where only one service has persistent anomalies.

5. Publishing Metrics

- Flowchart Step: Outputs Sev 1 and Sev 2 incidents as binary flags.
- **My Code Enhancements:** Publishes Sev 1 and Sev 2 metrics to Prometheus:

```
sev1_gauge.set(sev1_flag)
sev2_gauge.set(sev2_flag)
```

Reason: Publishing these metrics allows integration with Grafana dashboards and other alerting systems, making it easy to track incidents visually.

6. Debugging and Resilience

- **Flowchart Step:** The flowchart assumes ideal conditions for detecting anomalies and processing metrics.
- **My Code Enhancements:** Includes retry logic for Prometheus queries with exponential backoff:

```
for attempt in range(retries):
    try:
        response = requests.get(PROMETHEUS_URL, params={"query": query}, timeout=15)
        ...
    except requests.exceptions.RequestException as e:
        time.sleep(2 ** attempt + random.uniform(0, 1))
```

    - Adds detailed logging for anomalies and incident status:

```
print(f"Accumulators: {accumulators}")
print(f"Temperature: {total_temperature}")
print(f"Severity Levels - Sev1: {sev1_flag}, Sev2: {sev2_flag}")
```

    - Reason:
        - These enhancements improve the resilience of the code against network issues and missing data in Prometheus.
        - Debugging output provides insights during testing and troubleshooting.

Why These Changes Improve the Flowchart

- Dynamic Configuration:
    - The flowchart assumes static service names and thresholds, but my code dynamically adapts to different configurations using command-line arguments (--services, --threshold, --port).
- Transient Handling:
    - By decrementing accumulators by 2 when no anomalies are detected, the code ensures transient issues do not trigger false positives, a refinement not explicitly shown in the flowchart.
- Resilience:

    - The retry mechanism and error handling make the system more robust in production environments where network or data issues may occur.
- Real-Time Monitoring:
    - Publishing metrics (incident_temperature, sev1, sev2) to Prometheus enables integration with Grafana, improving visibility and usability.

Console log of running incident detector with custom load with no anomaly

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl logs -f incident-detector-599bbf4f64-gvnfp
Starting Incident Detector on port 8003...
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
Error during Prometheus query for lab7_frontend_2_productcatalog: HTTPConnectionPool(host='prometheus.istio-system', port=9090): Max retries exceeded with url: /api/
ry?query=sum%28lab7_frontend_2_productcatalog_anomaly_count%29+by+%28job%29 (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x79c5847e1f4
iled to establish a new connection: [Errno 111] Connection refused'))
Retrying query for lab7_frontend_2_productcatalog: Attempt 2
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843418.828, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843418.84, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 0}
Temperature: 0
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843478.907, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843478.915, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 0}
Temperature: 0
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843538.989, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733843538.996, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 0}
```

Lab Task 4 – Detect Istio-generated raw anomalies and triggered incidents.

```
Starting Incident Detector on port 8003...
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
Error during Prometheus query for lab7_frontend_2_productcatalog: HTTPConnectionPool(host='prometheus.istio-system', port=9090): Max ret
ry?query=sum%28lab7_frontend_2_productcatalog_anomaly_count%29+by+%28job%29 (Caused by NewConnectionError('<urllib3.connection.HTTPConne
iled to establish a new connection: [Errno 111] Connection refused'))
Retrying query for lab7_frontend_2_productcatalog: Attempt 2
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733847259.804, '1']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733847259.817, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 1, 'lab7_frontend_2_shippingservice': 1}
Temperature: 2
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733847319.887, '1']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733847319.896, '1']}]
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl apply -f frontend_shipping_delay.yaml
virtualservice.networking.istio.io/shippingservice created
oadewusi@oluSec:~/aiops/Assignments/LAB7$ kubectl apply -f frontend_productcatalog_delay.yaml
virtualservice.networking.istio.io/productcatalogservice created
```

Applying one fault (frontend to shipping to get Sev1)

```
--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 2}
Temperature: 2
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815340.025, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815340.032, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 3}
Temperature: 3
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815400.099, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815400.107, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 4}
Temperature: 4
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815460.174, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815460.184, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 5}
Temperature: 5
Severity Levels - Sev1: 0, Sev2: 1
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816060.917, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816060.924, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 6}
Temperature: 6
Severity Levels - Sev1: 0, Sev2: 1
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816120.992, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816120.999, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 4}
Temperature: 4
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------
```

Applying double fault to get Sev2.

```
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815640.404, '1']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815640.411, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 1, 'lab7_frontend_2_shippingservice': 8}
Temperature: 9
Severity Levels - Sev1: 1, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815700.478, '1']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815700.484, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 2, 'lab7_frontend_2_shippingservice': 9}
Temperature: 11
Severity Levels - Sev1: 1, Sev2: 0
----------------------------------

anomaly-monitor-frontend-productcatalog-677b894f5f-6fkrp
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815760.551, '1']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733815760.558, '1']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 3, 'lab7_frontend_2_shippingservice': 10}
Temperature: 13
Severity Levels - Sev1: 1, Sev2: 0
----------------------------------
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816000.845, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816000.853, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 1, 'lab7_frontend_2_shippingservice': 8}
Temperature: 9
Severity Levels - Sev1: 1, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816060.917, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733816060.924, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 6}
Temperature: 6
Severity Levels - Sev1: 0, Sev2: 1
```

Lab Task 5 – Generate transient anomalies and show them tolerated by incident detector

```
}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 1, 'lab7_frontend_2_shippingservice': 2}
Temperature: 3
Severity Levels - Sev1: 0, Sev2: 0
----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819406.908, '1']}
]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819406.917, '1']
}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 2, 'lab7_frontend_2_shippingservice': 3}
Temperature: 5
Severity Levels - Sev1: 1, Sev2: 0
```

Console logs showing transient generation and associated anomaly and incident detection.

```
Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819406.908, '1']}
]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819406.917, '1']
}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 2, 'lab7_frontend_2_shippingservice': 3}
Temperature: 5
Severity Levels - Sev1: 1, Sev2: 0
-----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819466.992, '0']}
]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819466.999, '0']
}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 1}
Temperature: 1
Severity Levels - Sev1: 0, Sev2: 0
-----------------------------------
```

```
--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 2, 'lab7_frontend_2_shippingservice': 3}
Temperature: 5
Severity Levels - Sev1: 1, Sev2: 0
-----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819466.992, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819466.999, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 1}
Temperature: 1
Severity Levels - Sev1: 0, Sev2: 0
-----------------------------------

Querying Prometheus for lab7_frontend_2_productcatalog: sum(lab7_frontend_2_productcatalog_anomaly_count) by (job)
[lab7_frontend_2_productcatalog] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819527.067, '0']}]
Querying Prometheus for lab7_frontend_2_shippingservice: sum(lab7_frontend_2_shippingservice_anomaly_count) by (job)
[lab7_frontend_2_shippingservice] Query result: [{'metric': {'job': 'kubernetes-pods'}, 'value': [1733819527.076, '0']}]

--- Incident Monitoring Status ---
Accumulators: {'lab7_frontend_2_productcatalog': 0, 'lab7_frontend_2_shippingservice': 0}
Temperature: 0
Severity Levels - Sev1: 0, Sev2: 0
-----------------------------------
```

Explanation of How Transients Were Configured and Why

Transient Configuration

The transient load configuration is implemented in the transient_in_effect function within the provided file.

1. **Transient Start and End Times**:
   - **transient_min_run_secs**: The minimum runtime (in seconds) when a transient spike begins. Configured as [30, 600].
   - **transient_max_run_secs**: The maximum runtime (in seconds) when a transient spike ends. Configured as [150, 720].

Why:

- These values simulate two transient periods within the total test duration.
- The first transient starts at 30 seconds and lasts for 120 seconds.

   - The second transient starts at 10 minutes (600 seconds) and lasts for 2 minutes (120 seconds). This mimics realistic temporary traffic surges, such as user spikes during promotions or high-demand periods.
2. **Transient Surge in Users**:
   - **transient_surge**: Configured as [1000, 1000].
   - During each transient period, 1000 additional users are added to the traffic.

Why:
   - The transient surge was chosen to test the resilience of the incident detection system against large but temporary traffic increases. This ensures that short-lived anomalies do not unnecessarily trigger incidents.
3. **Behavior of transient_in_effect**:
   - The function checks if the current runtime (run_secs) is within any of the defined transient periods. If true, it returns the corresponding transient_surge value to add a temporary spike in users.
   - **Example**: If run_secs is 45, it adds a transient spike of 1000 users (first transient period).

The transients are integrated into the overall load test shape using the MyCustomShape class.

1. **Key Parameters**:
   - **time_limit**: Total runtime of the load test. Default: 3600 seconds (1 hour).
   - **spawn_rate**: Number of users spawned per second. Default: 100.
   - **max_users**: Maximum number of users. Default: 250.
   - **cycle_time**: Total duration for a complete up-and-down traffic cycle. Default: 1200 seconds (20 minutes).
   - **num_steps**: Number of steps for the up-and-down cycle. Default: 10.
2. **Transient Integration**:
   - In each step of the traffic cycle (tick function), the transient_in_effect function is called to check if a transient spike should be applied.
   - The transient spike (transient_surge) is added to the target user count:

   ```
   return (target_users + transient_spike, self.spawn_rate)
   ```
3. **Why**:
   - This ensures that transients are applied on top of the regular traffic cycle, making them distinct and noticeable in the incident detector's metrics.
4. **Example Transient Configuration in the Cycle**:
   - Suppose time_limit=900 seconds, cycle_time=300 seconds, and num_steps=5:
     - Regular traffic gradually increases and decreases every 60 seconds.
     - Transient spikes (1000 users) occur during the specified periods, creating sharp, temporary peaks.

Why These Configurations Were Chosen

1. **Realistic Traffic Spikes**:
   - The transient configuration mimics real-world scenarios like promotional traffic surges or sudden user activity increases.
2. **Testing Incident Detector Tolerance**:
   - The transient periods and user surges were chosen to test whether the incident detector can distinguish between transient anomalies and persistent issues.
   - The decrement-by-2 logic in the incident detector ensures that short-lived transients do not cross the incident threshold, validating the system’s design.
3. **Integration with Regular Traffic**:
   - By overlaying transients on a cyclic traffic pattern, the configuration tests the detector’s ability to handle transients within varying baseline loads.

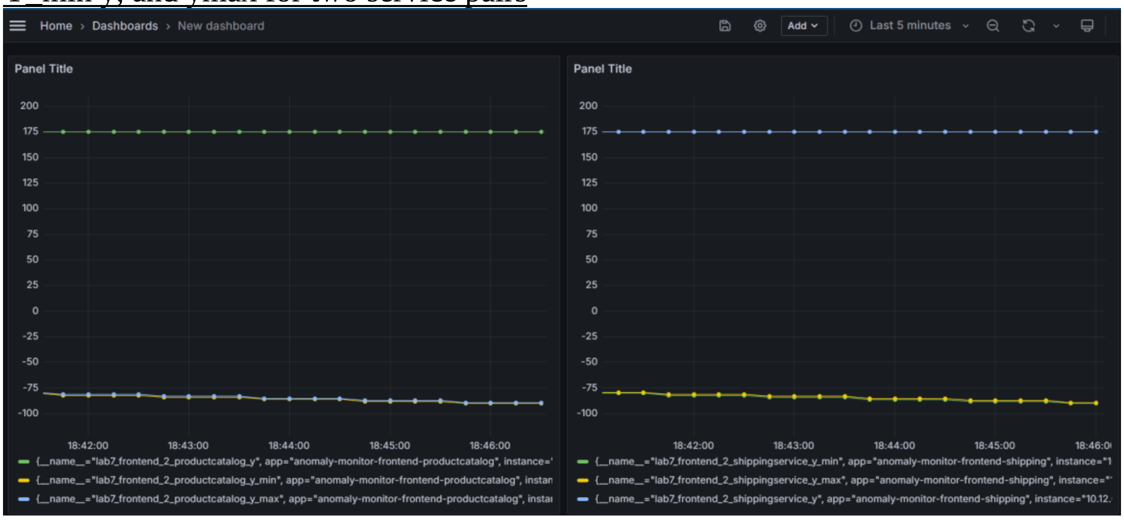
Lab Task 6 – Design Grafana dashboard for raw request times, anomalies, and incidents

```
histogram_quantile(0.5,
rate(istio_request_duration_milliseconds_bucket{source_app="frontend",
destination_app="shippingservice"}[1m]))
```

AND

```
histogram_quantile(0.5,
rate(istio_request_duration_milliseconds_bucket{source_app="frontend",
destination_app="productcatalogservice"}[1m]))
```

Y_min y, and ymax for two service pairs



incident_sev1 and incident_sev2